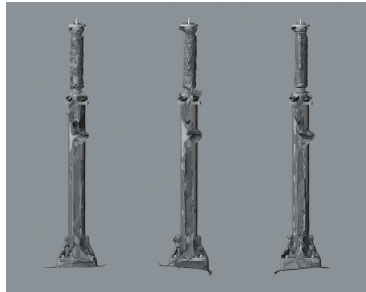


Why the Reconstruction Loop Broadens: Three Structural Transitions

Each row shows how a new dependency changes the engineering decision, not just the fitted parameters.

A. Local fit → reusable scope

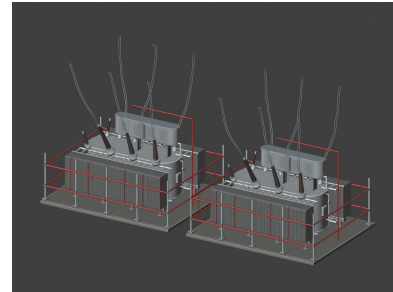
B01: local fit



define a local comparison domain, then fit pose and size

repetition introduces invariance

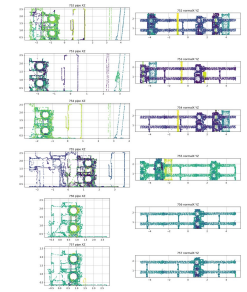
B08: shared equipment



two installations reuse one shared MASTER

reuse boundary becomes testable

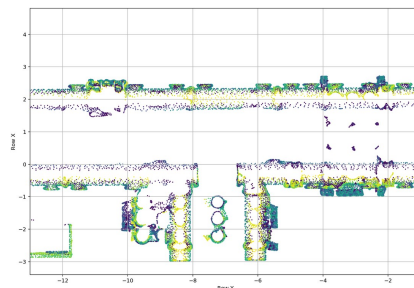
B15: revise reuse boundary



finite site geometry moves outside the shared asset

B. Object geometry → connected representation

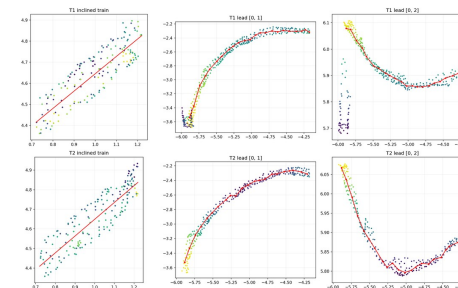
B20: ports and routes



existing equipment ends become explicit ports

relationship constraints appear

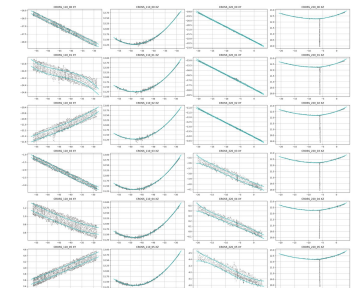
B23: change representation



the connection is reformulated from straight to curved

endpoint / topology constraints scale

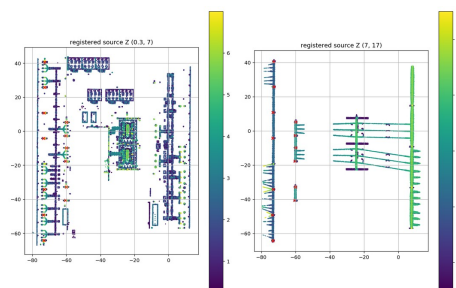
B29: relational constraints



continuity and endpoint constraints scale across subsystems

C. Scheduled addition → gap/state-driven closure

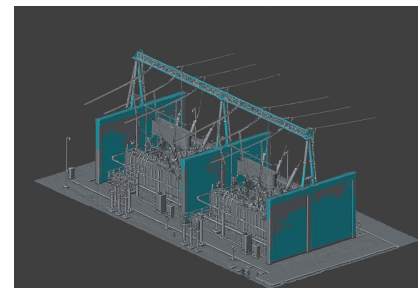
B25: unexplained geometry



large unexplained structures remain in the scene

coverage becomes a task signal

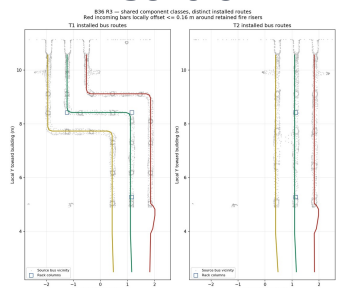
B26: reconstruct selected gaps



coverage drives which missing structures to reconstruct next

new geometry must fit the existing world

B36: state-constrained insertion



new geometry must fit inherited endpoints and occupied space

Descriptive evidence from the B01-B36 record; the figure explains the changing reconstruction problem rather than proposing a universal complexity taxonomy.